

Log Intrusion Detection System (1).docx

by Yogja Tiwari 220953534

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Log Intrusion Detection System

¹ Ankit Mishra
Department of ICT
Manipal Institute of Technology
Manipal-576104, Karnataka
220953658

¹ Ankit Tojo
Department of ICT
Manipal Institute of Technology
Manipal-576104, Karnataka
220953486

¹ Yogja Tiwari
Department of ICT
Manipal Institute of Technology
Manipal-576104, Karnataka
220953534

1.Introduction

In the digital era, the increasing reliance on online services and remote access solutions like OpenSSH has brought with it significant security challenges. OpenSSH, while crucial for secure remote connections, is frequently targeted by malicious actors aiming to exploit vulnerabilities and gain unauthorized access. Intrusions can take many forms, such as brute-force login attempts, unauthorized IP access, and compromised credentials. Without adequate monitoring, these threats often go undetected until substantial damage has been done, risking sensitive information and impacting system integrity. As attackers employ more sophisticated methods to bypass security protocols, organizations are compelled to adopt advanced tools that not only detect such unauthorized access attempts but also allow them to quickly and effectively mitigate these risks. Timely detection of these potential threats is crucial, as even a slight delay can have catastrophic effects on an organization's infrastructure and data security.

This Log Intrusion Detection System (LIDS) for OpenSSH addresses these critical security challenges by continuously monitoring and analyzing OpenSSH system logs to identify signs of suspicious or unauthorized activities. The system leverages advanced logging mechanisms, integrating real-time detection and alerting functionalities that immediately notify administrators of potential intrusions. Strong authentication methods, such as OTP verification

and IP validation, ensure that only registered users can access the system, reducing the likelihood of unauthorized access. Additionally, encryption protocols protect the confidentiality and integrity of logs, making it challenging for intruders to tamper with evidence of their attempts. By implementing this OpenSSH log IDS, organizations can proactively secure their infrastructure, respond swiftly to potential threats, and maintain a high level of security for their critical assets.

2.Related Works

There are 5 types of Intrusion Detection Systems (IDS), and they are as follows:

- Network Intrusion Detection System(NIDS) - Set up at a planned point in the network and observe the passing traffic to match it to known attacks.
- Host Intrusion Detection System (HIDS) - monitors the incoming and outgoing packets for the local device it is on and alerts the administrator in case of malicious activity
- Protocol-Based Intrusion Detection System(PIDS) - This consists of a system set up at the front end of a server to monitor the protocols followed between a client and its server.
- Application Protocol-Based Intrusion Detection System(APIIDS) - Resides within a group of servers and identifies intrusions by monitoring communication on application specific protocols.

- 2 • **Hybrid Intrusion Detection System** - made by combination of two or more of the IDS listed above.

3.Features

Our current Intrusion Detection System has the following features:-

1. Log Sources: The system collects logs from different sources such as OpenSSH (for remote access logs), system authentication logs, and other system logs to have a comprehensive view of all activities happening within the system.

2. Log Collector: The logs are aggregated and stored in a text file which ensures that all log data is centralized for further analysis. The logs are encrypted to ensure that no unauthorized user can access them.

3. Log Analyzer: A log analysis component inspects the collected logs, detecting patterns and anomalies based on predefined rules and behavior analysis. This helps identify unauthorized access or irregular activities.

4. Intrusion Detection System (IDS): Once the logs are analyzed, any identified threats or suspicious activities trigger the IDS, which continuously monitors the system for potential attacks. This component can detect brute-force attacks, repeated failed login attempts, or unauthorized access attempts.

5. Audit Logs: The system generates detailed audit logs for compliance and regulatory purposes, allowing security teams to review past actions and logs to determine how an incident occurred.

6. Authentication System: The system follows a 2 factor authentication system when a new user is registered. An otp is sent to the user's email every time the user attempts to login.

4.Method

The program starts with a main menu where users have three options: "User Registration," "User

Login," and "Exit." This initial interface allows users to easily select the desired action, whether they are registering for the first time, logging in to access system features, or choosing to exit the application.

In the user registration process, the user is prompted to enter a username, Gmail address, and password. The application verifies that the provided email address is valid by ensuring it follows the format for a Gmail account (i.e., ending with "@gmail.com"). This email format validation is essential, as the email will later be used to send OTPs during login for enhanced security. If the email is valid, the password entered by the user is encrypted using bcrypt, a powerful hashing algorithm that includes a salt for extra protection against dictionary attacks. The encrypted password and email are then stored securely in a dictionary associated with the username. If a user attempts to register with a username that is already taken, they are notified and prompted to use a different one.

When a user chooses User Login, the system requires them to enter their username and password. The application then checks if the provided password matches the stored, encrypted password using bcrypt's verification function. If the password is correct, the system generates a One-Time Password (OTP) as an additional layer of security, sending it to the user's registered Gmail address. This OTP feature utilizes the Gmail SMTP server, requiring the system to authenticate using stored credentials and securely send the OTP email with a unique numeric code. The user then needs to enter the OTP they received to complete the login process. If the OTP entered matches, the user is successfully logged in and gains access to the system's main functionalities. If the OTP is incorrect, the login fails, and the user is notified.

After logging in, users are presented with a personalized User Login Page where they have three options: "Detect Log Intrusions," "Geotrack IP Addresses," and "Logout."

1. Detect Log Intrusions: In this feature, users are prompted to enter the path to an SSH log file they wish to analyze. The application then scans the file for multiple failed login attempts, focusing on repeated failed passwords associated with the user's username. It uses a timestamp-based system to identify sequences of failed attempts within a specific time frame (e.g., within five minutes). If it detects a pattern suggesting multiple failed login attempts over a short period, it triggers an alert, indicating potential intrusion. This feature provides users with a report of suspicious activities, listing the times and frequencies of these attempts to help them determine if further action is needed.

2. Geotrack IP Addresses: This feature enables users to track the origin of IP addresses found within the log file. After selecting this option, users enter the path to a log file, which the system then processes to extract and validate all IP addresses within it. Valid IP addresses are sent to the IPStack API for geolocation, and the application retrieves details such as the country and city of origin for each IP address. This information allows users to identify whether login attempts originated from familiar or unexpected locations, which can be a critical factor in detecting suspicious activity. If an IP address is found to be from a location that the user finds unusual or concerning, they can take further action to secure their system.

3. Logout: This option allows users to log out of their session and return to the main menu. Logging out ensures that users do not leave their session active, which could pose a security risk in a shared or public environment.

The process of scanning and validating IP addresses involves extracting all IPs from the log file and categorizing them as valid or invalid based on format. If a valid IP address is detected, the system checks the frequency of its occurrences within the log, which is useful for analyzing patterns in login attempts. The program then displays the counts for valid and invalid IPs and provides a report listing the valid IPs along with their frequencies.

The program includes robust error handling throughout, ensuring that users receive informative messages when issues arise. For instance, if the log file path provided does not exist or cannot be accessed, the system will notify the user with a file-not-found error. Similarly, if there are issues with the IP geolocation API (such as connectivity problems or response errors), the system will inform the user accordingly. These features help users navigate the program smoothly, even in cases where technical issues might otherwise disrupt the process.

The encryption, OTP validation, IP analysis, and intrusion detection functions are integrated seamlessly to create a holistic approach to login security. With each feature, the system aims to safeguard the user's information and provide visibility into potential threats. The multi-step login process with OTP verification strengthens authentication, while the log analysis tools offer insights into login activities and suspicious behaviors.

Overall, the Log Intrusion Detection System (LIDS) combines user-friendly design with advanced security practices, offering users the ability to register securely, authenticate with OTP, and monitor log files for signs of intrusion or irregular login attempts. Upon logout or exit, the application displays a thank-you message, marking the end of the user's session with an acknowledgment for using the system.

5. Output

```
---- LOG INTRUSION DETECTION SYSTEM ----
1. User Registration
2. User Login
3. Exit
Select an option: 1
Enter your email: karmaboy0017@gmail.com
Enter your password:
User has been registered successfully!

---- LOG INTRUSION DETECTION SYSTEM ----
1. User Registration
2. User Login
3. Exit
Select an option: 2
Enter your name: yogja
Enter your password:
Generating OTP...
OTP sent successfully to karmaboy0017@gmail.com
Enter the OTP sent to your email: 527841
Login successful! Welcome, yogja.
```

Fig1: User registration and login

```
1. Detect log intrusions
2. Geotrack IP addresses
3. Logout
Select an option: 1
Enter the path of the log file: EventLogs.txt
Analyzing event logs...
Loading data...

[2024-12-31 21:03:10] Multiple failed login attempts for user "yogja" within 5 minutes at 2024-12-31 21:03:10 and 2024-12-31 21:03:12.

---- 500 Log IP Validation Report ----
Total Visits processed: 36

Geotrack IP addresses:
-----
IP: 192.168.135.72 Count: 6
IP: 192.168.238.228 Count: 6
IP: 192.168.238.207 Count: 1
IP: 8.8.8.8 Count: 1
IP: 198.52.3.4 Count: 1
```

Fig2: Intrusion Detection

```
1. Detect log intrusions
2. Geotrack IP addresses
3. Logout
Select an option: 2
Enter the path of the log file: EventLogs.txt
Analyzing IP addresses...
IP: 192.168.238.228 | Country: None | City: None | Visits: 6
IP: 192.168.238.207 | Country: None | City: None | Visits: 1
IP: 192.168.135.72 | Country: None | City: None | Visits: 6
IP: 8.8.8.8 | Country: United States | City: Glenmont | Visits: 1
IP: 198.52.3.4 | Country: United States | City: Staples | Visits: 1
```

Fig 3: Geotracking IPs

6. Conclusion

The logs generated on the servers are extremely vital for monitoring any suspicious activity that there may be. By making use of the logs and analyzing them, we can accurately determine an attack in real-time and take steps to remediate it. Firewalls and authentications are effective in preventing unauthorized access, but fail in

monitoring of network traffic. This necessitates the need for an effective traffic monitoring system. IDS provides us an easy to use, flexible and robust method to monitor said traffic and search for any suspicious activity or attacks.

7. Reference

[1]<https://www.slideshare.net/slideshow/intrusion-detection-system-project-report/135838221#8>

[2]https://www.researchgate.net/publication/342353650_Log_Based_Intrusion_Detection_System

[3]Youakim Badr. " Enabling intrusion detection systems with dueling double deep -learning " , Digital Transformation and Society, 2022

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